

SEUPD@CLEF: Team RETRIX on Source Retrieval for Scientific Web Claims

Notebook for the CheckThat! Lab at CLEF 2026

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Abstract

We present the results of the RETRIX group for the CheckThat! Lab at CLEF 2026 Task 1, whose goal is to identify the scientific paper implicitly referenced in a social media post. The input consists of short, noisy and usually informal posts that mention scientific findings without providing explicit links or citations. Our pipeline first translates non-English posts into English, then expands them using an embedding model and terms extracted from the top five documents ranked by similarity. After that it is applied a hybrid retrieval system that combines dense FAISS search with sparse lexical matching and ColBERT multi-vector scoring ($s = 0.5 s_{\text{dense}} + 0.15 s_{\text{sparse}} + 0.35 s_{\text{colbert}}$). Finally, we perform neural reranking with a fine-tuned cross-encoder model.

Keywords

Fact Checking, Information Retrieval, Scientific Document Retrieval, Social media fact verification, CLEF, Neural Re-ranking, Cross Encoder

1. Introduction

The rapid spread of scientific claims through social media creates a difficult fact-checking scenario: posts often report, simplify or sensationalize research findings while omitting the original source. In many cases, the cited study is not linked explicitly with a URL or a bibliographic reference but only mentioned indirectly through fragments such as the reported finding, the venue, the institution or the authors. As a consequence, verifying such claims requires first identifying the scientific publication that is being implicitly referenced.

This problem is the focus of Task 1 of the CLEF 2026 CheckThat! Lab, *Source Retrieval for Scientific Web Claims*: given a social media post containing a scientific claim and an implicit reference to a paper, the goal is to retrieve the correct publication from a fixed collection of 10,000 candidate papers written in English.

The task is challenging for several reasons. First, the linguistic gap between social media and scientific writing is substantial: posts are short, informal and often contain paraphrases, slogans, hashtags or incomplete statements whereas scientific papers use more precise and technical language. Second, the reference to the source paper is implicit rather than explicit so the retrieval system must infer the connection from partial semantic evidence rather than from direct citations. Third, relevant clues may appear in different forms, such as mentions of a scientific result, a medical outcome, a research institution or a publication venue, requiring the system to combine lexical and semantic evidence effectively.

An additional challenge is multilinguality: the official query sets include English, German and French posts while the candidate pool remains shared across languages.

“Search Engines”, course at the master degree in “Computer Engineering”, Department of Information Engineering, and at the master degree in “Data Science”, Department of Mathematics “Tullio Levi-Civita”, University of Padua, Italy. Academic Year 2025/2026

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Team RETRIX approaches this problem with a four-stage sequential pipeline.

1. In a preliminary stage we translate to English all the queries written in French and German using the model `utter-project/EuroLLM-9B-Instruct-2512` [1], specifically funded by the EU for understanding and generating text in all European languages. This allows us to handle all queries the same way, independently of the original language and consequently of the semantic and structural differences.
2. Raw tweet-style queries are cleaned of emojis and symbols like "#" and "@" and then enriched via pseudo-relevance feedback: `BAAI/bge-large-en-v1.5` [2] encodes both queries and corpus documents into a shared embedding space, retrieves the top-5 semantically similar papers per query and extracts the most frequent terms from those papers as expansion tokens.
3. The expanded queries are fed into `BAAI/bge-m3` [3] operating in hybrid mode, which combines dense inner-product search via FAISS [4] with sparse lexical matching and ColBERT multi-vector scoring to produce an initial ranked list of candidates.
4. Finally, the top candidates are re-ranked by a neural cross-encoder, in our case `nvidia/llama-nemotron-rerank-1b-v2` [5], fine-tuned on the English training dataset of the CLEF CheckThat! Task 1 2026 and all the queries' datasets of the 2025 CLEF CheckThat! SubTask 4b.

The paper is organized as follows: Section 2 discusses the most relevant research directions related to this task; Section 3 describes in more detail our approach; Section 4 explains our experimental setup; Section 5 discusses our main findings; finally, Section 6 draws some conclusions and outlooks for future work.

2. Related Work

A starting point is lexical retrieval where probabilistic ranking models such as BM25 remain strong baselines thanks to their efficiency, interpretability and robustness in candidate generation [6]. In the setting of scientific claim source retrieval, lexical matching is especially useful when a post mentions recognizable entities or terminology that also appears in the title, abstract, venue or author list of the referenced paper.

Neural retrieval methods have been proposed to mitigate the lexical mismatch between queries and documents by learning dense semantic representations. More broadly, neural information retrieval has shifted attention from exact token overlap toward representation learning and learned matching functions [7]. Within this line of work, dense dual-encoder architectures such as Dense Passage Retrieval (DPR) demonstrated that learned query and document embeddings can be highly effective for first-stage retrieval [8]. This is especially relevant to our task because social media posts often paraphrase scientific findings rather than reuse the exact wording found in the original publication. Building on this direction, BGE-M3 unified dense, sparse, and multi-vector retrieval in a single multilingual model [3], enabling hybrid scoring that combines embedding similarity with lexical token weights, a combination particularly well-suited to cross-lingual settings where queries and documents share few surface tokens.

Multilingual information retrieval presents additional challenges beyond monolingual settings, as systems must bridge language mismatches before relevance can be estimated. Machine translation provides one practical solution, but it can introduce errors and vocabulary drift, especially for technical or domain-specific terminology [9].

Vocabulary mismatch between short, informal queries and longer scientific documents can be further reduced through query expansion. Classical pseudo-relevance feedback methods such as RM3 [10] address this issue by re-weighting query terms using top-ranked documents from an initial retrieval pass. Neural extensions of this paradigm replace or complement lexical evidence with dense representations [11]. Following this general approach, our pipeline incorporates embedding-based query expansion to enrich query representations before scoring against the corpus.

Between initial retrieval and final ranking, many modern systems adopt multi-stage pipelines to balance efficiency and effectiveness [7]. Within these pipelines, late-interaction models such as ColBERT

preserve fine-grained token-level matching while remaining substantially more scalable than fully pairwise neural scoring over the whole collection [12]. Similarly, transformer-based re-rankers have become a common second-stage component for refining an initial candidate set through stronger query-document interactions [13]. Taken together, these approaches motivate retrieval architectures that first generate a manageable set of candidates and then apply more expensive relevance estimation only where it is most useful.

Recent work has argued that reranker fine-tuning benefits from hard negatives because they force the model to learn finer-grained relevance boundaries than easy or random negatives provide [14]. Adapting large language models to retrieval tasks is further facilitated by parameter-efficient fine-tuning with LoRA [15], which has been shown to match or approach full fine-tuning quality for passage reranking while substantially reducing GPU memory requirements [16]. This perspective is also consistent with recent practical guidance for training cross-encoder rerankers with Sentence Transformers [17]. In that setting, fast embedding-based retrieval is used to build a candidate pool, after which the cross-encoder jointly scores query-document pairs and benefits from negatives mined from nearby embedding neighborhoods rather than from trivial mismatches.

3. Methodology

We started from a lexical baseline based on Apache Lucene BM25 and progressively moved to a multi-stage neural pipeline.

The final pipeline (Figure 1) for the proposed information retrieval system, consists of four stages:

- **Query translation:** queries in French or German are first translated into English.
- **Query expansion:** queries are expanded with additional context.
- **Retrieval:** relevant documents are retrieved from the corpus using the expanded query.
- **Reranking:** the retrieved documents are reranked based on their relevance to the original query.

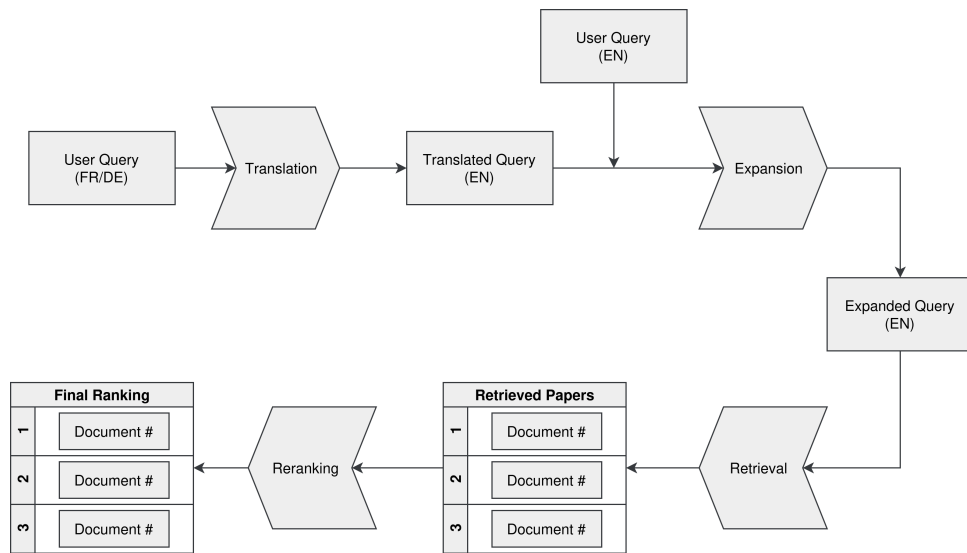


Figure 1: Four-stage pipeline for information retrieval.

3.1. Query translation

In the preliminary stage of our pipeline, non-English queries (French and German) are translated into English. This is done query by query using the model `utter-project/EuroLLM-9B-Instruct-2512` with the following prompt:

LLM Prompt

```
system: You are a professional translator.  
Translate the user's text to English.  
Output ONLY the English translation.  
Do not include the original text, explanations, labels, or any other content.  
  
user: Translate this {French/German} text to English.  
Output only the English translation, nothing else.  
  
{Query original text  $x$ }
```

Given x as the original query text, the model predicts probabilities for the next token with [18]:

$$P(y_t = i \mid y_{<t}, x) = \text{softmax}(z_t/T)_i$$

With our model configuration, the prediction becomes greedy, following [1]:

$$y_t = \arg \max_i P(y_t = i \mid y_{<t}, x)$$

The resulting English queries are then fed into the following step of our pipeline.

3.2. Query Expansion

Following the translation stage, each query is processed by a deterministic query-expansion module designed to reduce query–document vocabulary mismatch between short user claims and the terminology used in the scientific corpus. The module does not rely on generative LLM-based paraphrasing. Instead, it combines lexical feature extraction with dense semantic retrieval and neighborhood-based term induction.

- The translated English query is first analyzed with spaCy’s `en_core_web_sm` pipeline, which provides token-level linguistic annotations, including stop-word indicators and coarse-grained part-of-speech labels from the Universal POS tag set. Candidate lexical anchors are then identified by retaining tokens tagged as NOUN or PROPEN while excluding stop words and tokens shorter than three characters. The resulting set is deduplicated to produce a compact lexical signature that captures the principal nominal content of the query.
- The semantic expansion stage operates over a corpus in which each document is represented as the concatenation of its title and abstract. These title–abstract pairs are embedded with BAAI/bge-large-en-v1.5. In accordance with the documented BGE retrieval interface [2], query embeddings are produced by prepending the instruction `{Represent this sentence for searching relevant passages:}` to the translated query, whereas the corpus documents are encoded as retrieval targets. Similarity is then computed on normalized embeddings using the dot product. For unit-length vectors, dot-product scoring is equivalent to cosine similarity while being computationally more efficient.

$$s(q, d) = e_q^\top e_d = \sum_i e_{q,i} e_{d,i}$$

Because embeddings $e = \frac{v}{\|v\|_2}$ are L2-normalized $e_q^\top e_d = \cos(q, d)$ [19].

The five highest-scoring documents are retained as the local semantic neighborhood used for expansion.

In the described implementation, candidate expansion terms are extracted from the retrieved documents after lowercasing, ASCII normalization, mention’s and hashtag’s symbols removal, stop-word filtering and exclusion of short tokens (tokens ≤ 2 characters long are discarded). The candidate pool is aggregated across the retrieved neighborhood and assigned weights derived from the semantic similarity of the source documents so that terms originating from more similar

neighbors contribute more strongly. Terms already present in the query are removed and the fifteen highest-ranking candidates are retained as explicit expansion terms.

$$K = \text{unique}\{\ell(t) : POS(t) \in \{\text{NOUN}, \text{PROPN}\}, \neg stop(t), |t| > 2\}$$

$$w_h = \max(1, \text{round}(5s_h))$$

$$c(t) = \sum_h w_h tf(t, d_h)$$

The final expanded query is constructed by concatenating three components: the cleaned translated query, the noun and proper-noun-based keyword set extracted directly from the query and the expansion terms induced from the retrieved document neighbourhood. This output expanded query is then used by the downstream retrieval pipeline.

3.3. Retrieval

The retrieval stage is designed as a high-recall candidate-generation step between query expansion and neural reranking. Its objective is not to make the final relevance decision but to ensure that the true source paper is included in a sufficiently large candidate set for the subsequent, more computationally expensive cross-encoder stage. For this reason, after experimenting with a Lucene BM25 baseline, the final pipeline adopts **BAAI/bge-m3** in a hybrid retrieval setting.

The retriever searches the fixed paper collection provided by the task organizers. Each paper is represented at document level by concatenating its title and abstract (batch size = 1024) and the retrieved unit is always the paper pubkey. Although the collection includes metadata such as venue and authors, this hybrid retrieval stage uses only titles and abstracts to speed up the inference time.

We chose **BAAI/bge-m3** because it is well suited to this retrieval setting. The model can produce dense vectors, sparse lexical weights and ColBERT-style multi-vector representations from the same encoder.

This allows the retriever to combine broad semantic matching, exact term evidence and fine-grained token-level matching without switching between unrelated encoders.

For the dense representation, the document or query embedding is obtained from the normalized CLS representation [3]:

$$e = \text{norm}(H[0])$$

For the sparse representation, BGE-M3 assigns lexical weights to tokens [20]:

$$w_{qt} = \text{ReLU}(W_{lex}^\top H_q[i])$$

The sparse similarity between a query and a paper is then computed over overlapping terms:

$$s_{lex} = \sum_{t \in q \cap p} w_{qt} w_{pt}$$

Candidate selection is performed in two steps. First, all documents are encoded as dense BGE-M3 vectors and searched with an inner-product FAISS index [4]. The dense FAISS score is computed using inner product [21]:

$$s_{IP}(q, d) = q^\top d$$

Since the dense vectors are normalized, this inner product is equivalent to cosine similarity:

$$q^\top d = \cos(q, d)$$

This dense stage acts as an efficient broad filter over the whole collection. The default setting retrieves a large candidate pool of `candidate_multiplier × top_k`, with `candidate_multiplier = 3` and

`top_k` = 2000, before the final truncation. The number of candidates passed to the hybrid rescoring stage is:

$$K_c = \min(N, \max(K, Km))$$

This deliberately large candidate set favours recall, which is more important than precision at this stage.

Second, the dense candidates are rescored using the sparse and ColBERT-style representations. The sparse component rewards important overlapping terms, while the ColBERT-style component captures local token correspondences that may be lost when each document is represented by a single dense vector [12]. The ColBERT-style late-interaction score is computed as [12]:

$$s_{colbert}(q, d) = \frac{1}{N_q} \sum_{i=1}^{N_q} \max_{1 \leq j \leq N_d} q_i^\top d_j$$

The final configuration was selected through a series of development runs. We first tuned the lexical baseline by varying the Lucene analyzer, stop-word lists, stemming strategy and BM25 searcher variants. These experiments improved the classical baseline but also highlighted its limitations for this task: even with query expansion and reranking, BM25-based candidate generation remained more dependent on surface-level lexical overlap than desired.

We also tested BGE-M3 in a hybrid retrieval setting, using a setup that differed slightly from the final configuration. In those experiments, sparse, dense and ColBERT-style retrieval were performed using dedicated query expansions tailored to each retrieval mode. More specifically, each query was represented through three complementary views: an embedding view for semantic matching, a sparse view for lexical and term-based evidence, and a ColBERT view for fine-grained token-level interaction.

However, the best results in our analysis were obtained with a simpler configuration in which BGE-M3 hybrid retrieval used the same expanded query, described in the previous section, for all three retrieval modes. We then tested several weighting configurations. Among them, the run labelled `bge_m3_hybrid_513` was retained because it provided the strongest recall-oriented trade-off among the retrievers evaluated without reranking. This configuration used weights of 0.5, 0.15 and 0.35 for dense, sparse, and ColBERT scoring, respectively, with a multiplier of 3 and a maximum hybrid input length of 384.

A closely related configuration with a larger ColBERT weight achieved a nearly identical MRR@5 but a lower Recall@100. We therefore preferred the configuration that better matched the role of retrieval as a candidate generator. This choice was further confirmed by downstream experiments: reranking candidates from `bge_m3_hybrid_513` consistently outperformed the corresponding BM25-based reranked runs. In addition, increasing the reranked candidate depth from a few hundred documents to between 1000 and 2500 documents increased the final recall by giving the neural reranker a larger candidate pool.

The final hybrid retrieval score is therefore computed as:

$$s(q, d) = 0.5 s_{dense}(q, d) + 0.15 s_{sparse}(q, d) + 0.35 s_{colbert}(q, d).$$

The largest weight is assigned to the dense score because it provides the main semantic signal. The sparse and ColBERT components add complementary lexical and fine-grained matching evidence. Candidates are sorted by this score in descending order and truncated to the final `top_k`. The retrieval output is a JSON mapping from each query ID to an ordered list of paper pubkeys. Scores are not exported, because the next stage treats retrieval as a candidate generator and recomputes relevance using a cross-encoder over the returned document identifiers.

3.4. Reranking

3.4.1. Cross-encoder scoring

Our strongest single-stage reranker is `nvidia/llama-nemotron-rerank-1b-v2`, a 1B-parameter LLaMA-based model fine-tuned as a sequence classifier. For each query-document pair, we build an

input sequence using the official NVIDIA prompt template:

LLM Prompt

```
question:{query} \n \n passage:{passage}
```

The model computes a scalar logit $z \in \mathbb{R}$ through a linear classification head applied to the final hidden state and the relevance score is obtained by applying a sigmoid:

$$s(q, d) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Documents are then sorted in descending order of $s(q, d)$. In practice, reranking deeper candidate pools (500-2500 documents) consistently produced better final results than shallower pools, confirming the importance of high recall at the retrieval stage to compensate for miss-labeling from weaker previous retrieval-stages. At this stage, queries are evaluated against the corpus of the documents taking into account the title, abstract and author of the documents.

We also evaluated alternative rerankers, including Qwen/Qwen3-Reranker-4B [22] and Alibaba-NLP/gte-reranker-modernbert-base [23, 24] variants.

3.5. Fine-tuning the Reranker

To improve the Nemotron reranker on the domain-specific characteristics of the CheckThat! task, we fine-tuned `nvidia/llama-nemotron-rerank-1b-v2` using Parameter-Efficient Fine-Tuning (PEFT) with Low-Rank Adaptation (LoRA) on data from both the 2025 and 2026 editions of the task.

3.5.1. Training data and hard-negative mining

Training examples are query-document pairs with binary labels: a positive document is the gold source paper associated with the query, while negative documents are hard negatives mined using `sentence-transformers/static-retrieval-mrl-en-v1` [25, 26, 27]. Hard negatives are selected from the range $[5, 50]$ nearest neighbors (by embedding similarity) that satisfy score ≤ 0.85 and lie at least 0.05 below the positive score in cosine similarity (absolute margin) preventing trivially easy or falsely relevant negatives from polluting training that would under-mine the effectiveness of the fine-tuning.

Training data combines two years of the task:

- **2026 data:** split 90% training / 10% development. The development set is drawn exclusively from 2026 data for consistency with the official CT26 Task 1 evaluation protocol. Due to time-constraints we used just the `en_train` dataset but we suppose that also German and French translated queries datasets could improve the quality of the fine-tuning;
- **2025 data:** all available topics (dev, test-gold and train splits) are used entirely for training.

This design ensures that the dev set used to track MRR@5 during training is distribution-aligned with the contest test queries. Each training record contains one query, one positive passage and up to five hard negatives.

3.5.2. LoRA architecture

To fine-tune the 1B-parameter base model with limited GPU memory, we apply LoRA [15] with rank $r = 16$ and scaling factor $\alpha = 32$. Adapters are inserted into all attention and feed-forward projection layers: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj` and `down_proj`. The classification head (score) is included in `modules_to_save` and is fully updated during training. The task type is `SEQ_CLS`, matching the sequence-classification objective of the reranker.

3.5.3. Training objective

The model produces a single scalar logit z per input pair. Training minimizes binary cross-entropy with logits over all positive ($y = 1$) and negative ($y = 0$) pairs in each batch:

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B \left[y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \right]$$

where B is the effective batch size and σ is the sigmoid function. Gradient accumulation over 4 steps is used to simulate a larger effective batch size of 32 without additional memory overhead.

3.5.4. Optimization

The model is trained with AdamW, a learning rate of 2×10^{-5} , a linear warmup over 10% of the total training steps and gradient clipping at norm 1.0. Training runs for 3 epochs with bfloat16 mixed precision and gradient checkpointing enabled to reduce VRAM usage. The checkpoint with the best MRR@5 on the 2026 development set is retained.

3.5.5. Inference

At inference time, the LoRA adapter weights are merged into the base model via `merge_and_unload()` before scoring, so no PEFT overhead is incurred during evaluation. The scoring procedure and prompt format are identical to those of the base reranker (Section 3.4) and the fine-tuned model is used as a drop-in replacement in the reranking stage.

3.6. Evaluation

We evaluate systems on the CheckThat! Task 1 query sets by matching retrieved pubkeys against the gold pubkey for each query. The official evaluation metric for the task is MRR@5. We also tracked Recall@100 to monitor candidate-generation quality and to verify that reranking gains do not come from overly aggressive filtering.

4. Experimental Setup

All experiments were conducted on our own machines, using a computer with an AMD Ryzen 9 5900X, 32 GB of RAM and an NVIDIA RTX 3090 GPU (10,496 CUDA cores, 1.70 GHz boost clock, 24 GB of GDDR6X memory, 35.58 TFlops) for resource-intensive workloads and our laptops for standard workloads and program refinement. We used four laptops: the two most powerful are a MacBook Air M3 and a machine with no dedicated GPU, 16 GB of RAM and an Intel Core i7-13700H. For reproducibility, all environment settings have been documented and are available in the Bitbucket repository at <https://bitbucket.org/upd-dei-stud-prj/seupd2526-retrix/src/master/> or can be reproduced by following the instructions in this paper.

4.1. Dataset

The dataset consisted of 10,000 scientific publications that had to be paired with tweet-like posts written in three languages: German, French, and English. There are no explicit references such as URLs in any query. Instead, each tweet contains an implicit reference to a scientific paper, for example through mentions of authors or keywords related to the paper. The dataset files are all in JSON format. The scientific papers dataset contains the columns pubkey (unique identifier), title, abstract, venue/location and authors. The query datasets contain the columns index, pubkey (ID of the positive scientific paper that has to be matched with the query) and text.

Table 1
Dataset statistics by language

Split	Language	File	Queries		Length (tokens)			
			#	%	Min	Median	Max	Mean
Train	EN	en_train.json	14977	77.8	2	33	57	31.5
	FR	fr_train.json	2807	14.6	2	33	57	31.5
	DE	de_train.json	1460	7.6	2	33	57	31.5
Total			19244	100	2	33	57	31.5
Dev	EN	en_dev.json	3905	78.2	5	33	60	31.7
	FR	fr_dev.json	702	14.1	5	33	60	31.7
	DE	de_dev.json	386	7.7	5	33	60	31.7
Total			4993	100	5	33	60	31.7
Test	EN	final_en_test.json	6076	74.4	5	33	58	31.9
	FR	final_fr_test.json	1220	14.9	5	33	58	31.9
	DE	final_de_test.json	876	10.7	5	33	58	31.9
Total			8172	100	5	33	58	31.9
Overall Total			32409		2	33	60	31.7

Figure 2 compares lengths distribution of tokens across train, dev and test datasets. All three datasets share similar distributions, with the same median value and similar length value peaks.

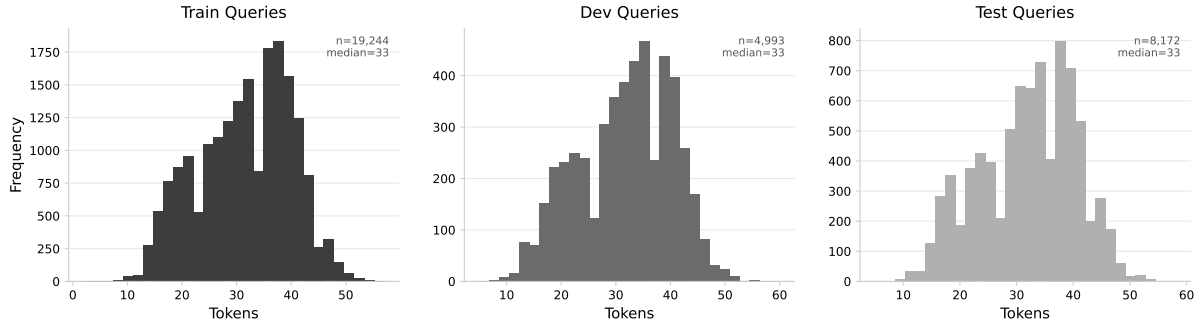


Figure 2: Train, dev and test datasets’ distribution of query token length.

5. Results and Discussion

5.1. Overview

This section summarizes the empirical evidence that supports our pipeline choices and diagnoses remaining failure modes. We organize the presentation by

- descriptive summaries of effectiveness on the labeled TRAIN and DEV splits,
- formal per-query inference that tests the key design decisions, and
- error and rank-bucket analyses that explain residual failures.

The TRAIN English split (14,977 labeled queries) is our primary source for per-query statistical analysis because of its size; DEV splits are used for multilingual validation and TEST for submission diagnostics. Table 2 reports representative runs spanning lexical tuning, hybrid retrieval, and neural reranking. Values are computed on the TRAIN split and rounded to three decimals. The pipeline pattern is consistent: hybrid candidates improve recall relative to tuned lexical baselines, and neural reranking substantially improves early-precision metrics such as MRR@5.

Table 2
Representative English run-analysis results performed on `en_train_set`.

Family	Run label	Key configuration	MRR@5	Recall@100
Analyzer tuning	Analyzer V2	BM25 retrieval with <code>MyEnglishAnalyzer_v2</code> , <code>Searcher V2</code> , and <code>BGE-V1</code> query expansion	0.503	0.796
Analyzer tuning	Analyzer V1.12	BM25 retrieval with <code>KStem</code> and an <code>EBSCO-host/MEDLINE/CINAHL</code> stoplist variant	0.450	0.783
Retrieval only	Hybrid A	<code>BGE-M3</code> hybrid retrieval with dense, sparse, and <code>ColBERT</code> weights of 0.35, 0.15, and 0.50	0.557	0.837
Retrieval only	Hybrid B	<code>BGE-M3</code> hybrid retrieval with dense, sparse, and <code>ColBERT</code> weights of 0.50, 0.15, and 0.35	0.557	0.839
Best model reranking	Nemotron top-100	<code>bge_m3_hybrid_513</code> candidates reranked with <code>nvidia/llama-nemotron-rerank-1b-v2</code>	0.685	0.839
Best model reranking	Nemotron top-1000	<code>bge_m3_hybrid_513</code> candidates reranked with <code>nvidia/llama-nemotron-rerank-1b-v2</code>	0.702	0.911
Alternative rerankers	Qwen3-4B top-100	<code>bge_m3_hybrid_513</code> candidates reranked with <code>Qwen3-reranker-4B</code>	0.662	0.839
Alternative rerankers	GTE-ModernBERT top-100	<code>bge_m3_hybrid_513</code> candidates reranked with <code>Alibaba-NLP/gte-reranker-modernbert-base</code>	0.652	0.852

5.2. Evaluation evidence and metrics

We combine three complementary sources of evidence:

- the large TRAIN English run-analysis split (14,977 labeled queries) for robust per-query inference;

- multilingual DEV splits for validation across languages;
- the official TEST set for diagnostic checks on the submitted runs.

Because the pipeline is staged, we align metrics with stage objectives: Recall@100 is the primary diagnostic for the retrieval (candidate-generation) stage, while MRR@5 is the primary objective for the reranking stage. We additionally report MAP and nDCG@10 to describe ranking quality beyond the early positions.

Per-query MRR@5 is zero unless the gold paper appears within the first five ranks and equals $1/r$ when the first relevant document appears at rank $r \leq 5$. Confidence intervals reported in descriptive tables use the normal approximation over per-query scores and are explicitly labeled as descriptive summaries rather than inferential tests.

5.3. Run comparison on the TRAIN English split

Table 3 summarizes the main runs used in the TRAIN analysis. The hybrid variants differ only in the linear combination of dense, sparse, and ColBERT-style BGE-M3 signals; H513 is the retained hybrid used as the candidate generator for reranking. Reranking experiments apply the Nemotron family to H513 candidates with increasing pool depth (top-100, top-500, top-1000). The Δ column in the table reports differences against H513 to make the pipeline effect explicit.

Table 3

Run comparison on the TRAIN English split with 14,977 queries. Values are computed from ranked outputs and the corresponding gold labels.

Run	Family	Configuration	MRR@5	Recall@100	MAP	nDCG@10	Δ MRR@5
H424	Hybrid	BGE-M3 weights 0.40/0.20/0.40	0.5067	0.8277	0.5201	0.5531	-0.0500
H514	Hybrid	BGE-M3 weights 0.50/0.10/0.40	0.5141	0.8327	0.5273	0.5595	-0.0426
H414	Hybrid	BGE-M3 weights 0.45/0.10/0.45	0.5171	0.8327	0.5302	0.5621	-0.0396
H513	Hybrid	BGE-M3 weights 0.50/0.15/0.35	0.5567	0.8392	0.5688	0.5956	0.0000
H315	Hybrid	BGE-M3 weights 0.35/0.15/0.50	0.5572	0.8365	0.5692	0.5957	+0.0005
N100	Reranking	H513 candidates reranked with Nemotron, top-100 retained	0.6847	0.8392	0.6909	0.7146	+0.1280
N500	Reranking	H513 candidates reranked with Nemotron, top-500 retained	0.7010	0.8986	0.7096	0.7342	+0.1443
N1000	Reranking	H513 candidates reranked with Nemotron, top-1000 retained	0.7018	0.9109	0.7111	0.7357	+0.1451

H424 is the weakest reported hybrid configuration, while H513 and H315 perform substantially better on early-precision metrics. H513 has an advantage in Recall@100 relative to H315, which motivates its selection as the candidate generator for the full pipeline: retrieval should prioritize coverage that the reranker can refine. Neural reranking (Nemotron) yields the largest improvements: moving from H513 to N1000 increases MRR@5 by roughly 0.145 and increases Recall@100 notably, showing that deeper reranking improves both precision and coverage. However, gains in MRR@5 beyond top-500 are marginal, suggesting diminishing returns for very deep pools.

As shown later in Section 5.6, formal non-parametric inference supports these conclusions: the Friedman omnibus rejects equality across systems and Holm-corrected pairwise Wilcoxon tests identify the same stage-aligned differences summarized in Table 8.

5.4. Multilingual DEV split validation

Table 4 reports the fine-tuned Nemotron LoRA validation on multilingual DEV splits. DEV results validate the pipeline across translated queries: French consistently shows the strongest DEV performance, English is close behind, and German shows wider intervals due to fewer labeled queries. DEV runs were not used to re-tune the TRAIN analysis pipelines; they are presented as external validation with appropriately wider uncertainty bands.

The development results suggest that the final pipeline generalizes reasonably across languages after translation, but not uniformly. French is the strongest development split, with MRR@5 of 0.7599

Table 4

Fine-tuned Nemotron LoRA validation on labeled development data. The final two columns report rank-bucket counts for strong successes and unresolved failures.

Split	Queries	MRR@5	95% CI	Recall@100	MAP	nDCG@10	Rank 1 / Missing
EN DEV	3905	0.7160	[0.7030, 0.7290]	0.9129	0.7243	0.7483	2574 / 241
FR DEV	702	0.7599	[0.7310, 0.7888]	0.9288	0.7672	0.7894	496 / 40
DE DEV	386	0.6736	[0.6305, 0.7168]	0.8860	0.6820	0.7080	237 / 29

and Recall@100 of 0.9288. German is the weakest, with MRR@5 of 0.6736 and the widest confidence interval. Rank-bucket counts corroborate these summaries: French has the largest fraction of rank-1 successes and the fewest missing cases (496/40), while German shows fewer rank-1 hits and a higher unresolved fraction relative to its query count (237/29). In practice this pattern suggests that the reranker generalizes well to translated queries in French, English remains competitive, and that German’s lower point estimates and wider intervals are plausibly driven by both smaller sample size and translation/terminology drift. These DEV findings validate the pipeline qualitatively and motivated the decision to use a deeper top-2000 pool in the final TEST submission to favor recall.

5.5. Official TEST set submission

The official TEST input used for submission consists of 8172 queries: 6076 in English, 1220 in French, and 876 in German. The submitted systems use the same four-stage pipeline described in Section 3: translate non-English queries into English, expand the query, retrieve candidates with BGE-M3 hybrid retrieval, and rerank with the fine-tuned Nemotron LoRA cross-encoder.

Table 5

Official TEST set results metrics. The aggregate row is a micro-average over all queries.

Language	MRR@5	Recall@100	MAP	nDCG@10
EN	0.7117	0.9235	0.7202	0.7469
FR	0.7400	0.9328	0.7479	0.7689
DE	0.6806	0.9155	0.6907	0.7213
Aggregate	0.7126	0.9240	0.7212	0.7474

Table 5 reports micro-averaged metrics per language. TEST metrics confirm the DEV pattern: French shows the highest aggregate MRR@5, English is competitive, and German shows lower scores and larger uncertainty. For the TEST submission we used a deeper reranking pool (top-2000) to favor recall for the final delivered runs; DEV experiments indicated modest but consistent recall gains when increasing the pool depth, so we opted for a deeper pool in TEST to maximize coverage of the gold document.

5.6. Statistical analysis

We separate descriptive summaries from formal hypothesis tests. Descriptive confidence intervals (Figure 3) summarize uncertainty around mean per-query scores but do not by themselves imply statistical significance. For inference we follow a stage-aligned strategy: test retrieval-stage decisions on Recall@100 (coverage), and test reranking-stage decisions on MRR@5 (early precision). For each metric we perform a Friedman omnibus test over repeated measures; when the omnibus rejects, we report pairwise Wilcoxon signed-rank tests with Holm correction to control family-wise error across all tested pairs [28, 29].

5.6.1. Descriptive statistics

Table 6 summarizes per-query MRR@5 for H513 and Nemotron reranking depths on the TRAIN split. Reranking shifts the entire distribution: the median increases from 0.5 (H513) to 1.0 for Nemotron runs

and the first quartile rises from 0.0 to 0.25, indicating that a substantially larger fraction of queries receive a non-zero early-precision score after reranking.

Table 6

Descriptive statistics of per-query MRR@5 on the TRAIN English split. Confidence intervals use the normal approximation over queries.

Run	Mean	Median	Std. dev.	Variance	Q1	Q3	95% CI
H513	0.5567	0.5	0.4614	0.2129	0.00	1.00	[0.5493, 0.5641]
N100	0.6847	1.0	0.4303	0.1852	0.25	1.00	[0.6778, 0.6916]
N500	0.7010	1.0	0.4229	0.1788	0.25	1.00	[0.6943, 0.7078]
N1000	0.7018	1.0	0.4223	0.1784	0.25	1.00	[0.6950, 0.7086]

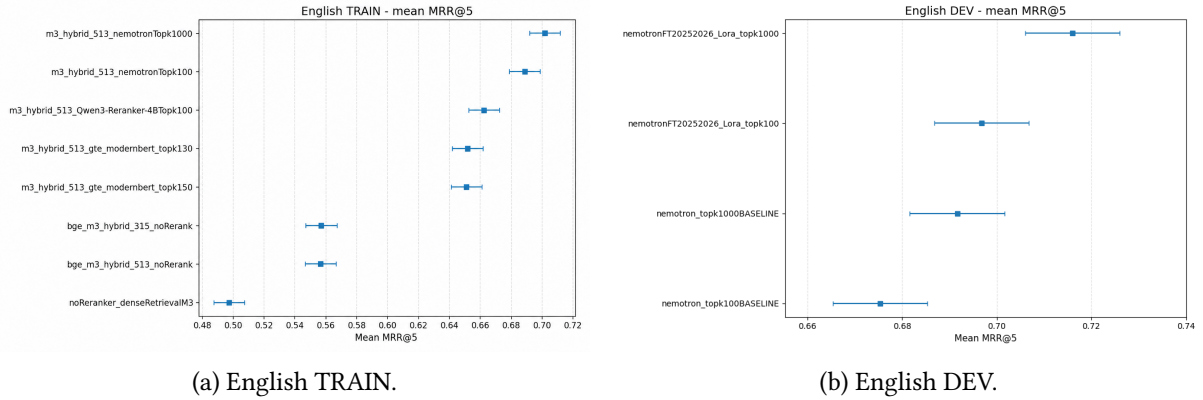


Figure 3: Mean MRR@5 with normal-approximation 95% confidence intervals for the English TRAIN and DEV statistical-analysis runs.

Figure 3 visualizes the mean per-query MRR@5 and the associated normal-approximation 95% confidence intervals for both the large TRAIN split and the smaller DEV split. The plot makes two descriptive points clear:

- Nemotron reranked systems lie well above the retained hybrid baselines on mean MRR@5, and
- the TRAIN CIs are much narrower than DEV’s (owing to its larger sample), which increases confidence in the descriptive means reported in Table 6.

These CIs are descriptive uncertainty bands (normal-approximation); formal hypothesis tests are reported in the following sections.

5.6.2. Formal significance testing

The Friedman omnibus test rejects the null hypothesis that all English TRAIN systems share the same per-query MRR@5 distribution; consequently we present Holm-corrected pairwise Wilcoxon tests to identify which differences are statistically supported. Table 7 shows the comparisons most relevant to pipeline decisions. Because of our staged pipeline, MRR@5 is the primary inferential target for reranking-stage comparisons, while Recall@100 is used for retrieval-stage comparisons.

The inference supports two practical conclusions. First, H513 and H315 are indistinguishable on MRR@5; choose between them using retrieval-stage diagnostics (Recall@100) and operational constraints. Second, the Nemotron reranker produces a large and statistically supported gain over the retained hybrid baseline. The small difference between N500 and N1000 is statistically detectable after Holm correction but practically negligible in MRR@5, suggesting practitioners may prefer N500 when latency or cost is a concern.

Table 7

Formal statistical tests on per-query MRR@5 for the English TRAIN split. The Friedman row is the omnibus test over all tested systems and 14,977 queries; pairwise rows use two-sided Wilcoxon signed-rank tests with Holm correction over all pairwise comparisons. Pairwise mean differences are computed as the second system minus the first.

Test	Systems / comparison	Mean difference	p -value	Interpretation
Friedman omnibus	All tested English TRAIN systems	–	< 0.001	At least one system differs
Wilcoxon–Holm	H315 vs H513	–0.0005	1.0000	No significant difference
Wilcoxon–Holm	H424 vs H513	+0.0500	8.58×10^{-116}	H513 significantly better
Wilcoxon–Holm	H513 vs N1000	+0.1451	< 0.001	N1000 significantly better
Wilcoxon–Holm	N500 vs N1000	+0.0008	0.0497	Significant but negligible in size

5.6.3. Stage-specific tests

To align inference with pipeline objectives we repeat Holm-corrected pairwise tests on Recall@100 for retrieval-stage comparisons and on MRR@5 for reranking-stage comparisons. Table 8 summarizes the comparisons that most directly motivated our pipeline choices.

Table 8

Holm-corrected pairwise tests aligned with the pipeline objective. For retrieval-stage comparisons, Δ is the preferred hybrid minus the comparator on Recall@100. For reranking-stage comparisons, Δ is N1000 minus the comparator on MRR@5.

Stage	Comparison	Δ	p_{holm}	Conclusion
Retrieval	H513 vs H315	+0.0026	0.0007	H513 better on Recall@100
Retrieval	H513 vs H414	+0.0065	0.0027	H513 better on Recall@100
Retrieval	H513 vs H424	+0.0115	7.81×10^{-9}	H513 better on Recall@100
Retrieval	H513 vs H514	+0.0065	0.0027	H513 better on Recall@100
Reranking	N1000 vs H513	+0.1451	< 0.001	N1000 better on MRR@5
Reranking	N1000 vs N500	+0.0008	0.0497	detectable but negligible

Together these tests support our pipeline choices. H513 is the best-supported hybrid candidate generator on Recall@100, and N1000 yields the largest early-precision gains on MRR@5. The incremental MRR@5 gain from N500 to N1000 is statistically detectable but practically tiny, which implies an operational trade-off between latency and marginal accuracy.

5.6.4. Rank-bucket and error analysis

Table 9 reports the rank position of the gold document for the retained hybrid and reranked runs. The analysis separates three qualitatively different error types. A gold paper is counted as *missing* only when its gold pubkey is absent from every document available in the ranked file for that query. Thus, for runs with retained deep candidate lists, missing cases indicate that the reranker cannot recover the document from its input pool. For runs available only as submitted top-100 lists, missing means absent from the available top-100 file, not necessarily absent from the original upstream retrieval pool. A gold paper appearing after rank 100 has been retrieved by the retained reranking pool but lies outside the Recall@100 cutoff. A gold paper appearing between ranks 6 and 100 is present in the candidate evidence but is ranked too low to satisfy the top-5 objective.

The labeled English comparison shows that the main effect of reranking is to move correct papers to the very top of the list. The best reranked run on the TRAIN split places the gold paper at rank 1 for 9697 queries, compared with 7459 for the retained hybrid single-stage run. It also reduces missing cases from 2409 to 929. The remaining errors are informative: for Nemotron Baseline, 1291 queries have the gold paper in ranks 11–100, indicating ranking failures rather than retrieval failures, whereas 929

Table 9

Rank buckets for the retained hybrid and reranked English runs. Percentages are computed within each split: 14,977 labeled English queries, 3,905 English DEV queries, and 6,076 English TEST queries.

Split	Family	Queries	Rank 1	Ranks 2–5	Ranks 6–10	Ranks 11–100	Rank > 100	Missing
EN TRAIN	Hybrid BGE-M3	14977	7459 (49.8%)	2280 (15.2%)	715 (4.8%)	2114 (14.1%)	0 (0.0%)	2409 (16.1%)
EN TRAIN	Nemotron Baseline	14977	9697 (64.7%)	2057 (13.7%)	598 (4.0%)	1291 (8.6%)	405 (2.7%)	929 (6.2%)
EN DEV	Nemotron LoRA fine-tuned	3905	2574 (65.9%)	538 (13.8%)	144 (3.7%)	309 (7.9%)	99 (2.5%)	241 (6.2%)
EN TEST	Nemotron LoRA fine-tuned	6076	3950 (65.0%)	926 (15.2%)	229 (3.8%)	506 (8.3%)	0 (0.0%)	465 (7.7%)

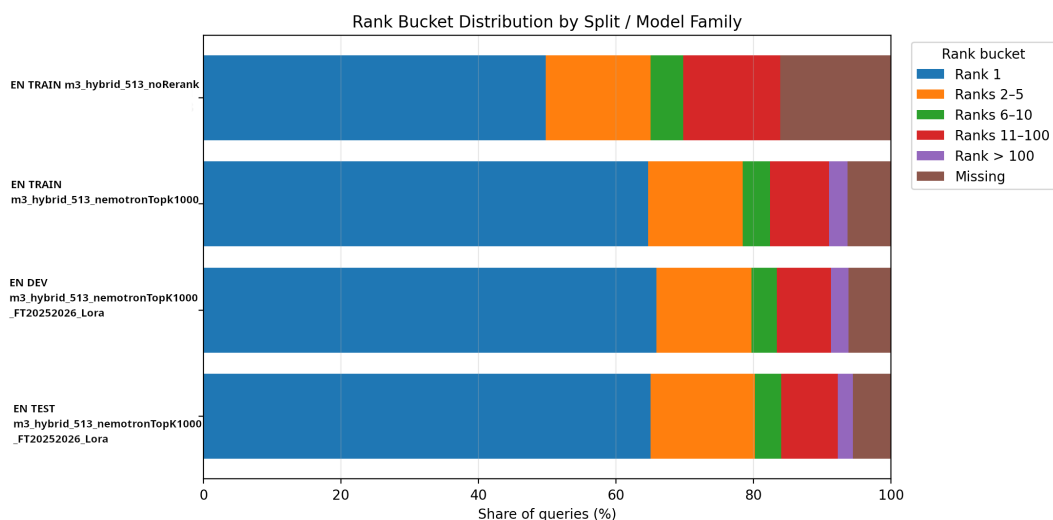


Figure 4: Visual representation of the rank bucket distribution for the retained hybrid and Nemotron runs on the EN TRAIN, DEV and TEST splits.

queries remain missing from the available reranking pool and therefore cannot be recovered by the reranker.

The final English DEV and TEST runs follow the same pattern. Nemotron LoRA fine-tuned places the gold paper at rank 1 for 2574 of 3905 queries, and does so also for the TEST set with 3950 of 6076 queries. Their top-5 coverage is similar, with 3112 DEV queries and 4876 TEST queries resolved within the first five ranks. The DEV file contains 1000 ranked documents for each query, so the 99 cases in Rank > 100 correspond to gold papers found by the retained reranking pool but not placed within the first 100 positions. The TEST file used to compute the metrics contains only the submitted top-100 documents per query; consequently, Rank > 100 is zero by construction, and the 465 missing TEST cases should be interpreted as missing from the available submitted run rather than as evidence that the original upstream retrieval pool never retrieved the gold paper.

Overall, the bucket analysis indicates that reranking substantially improves early precision while leaving two residual failure modes. The first consists of ranking failures, where the gold paper is present but remains below the top-5 cutoff. The second consists of missing cases, which are more severe because they cannot be repaired by reranking alone. These missing cases are likely associated with severe vocabulary mismatch, translation drift, or claims whose wording emphasizes entities and contextual cues that are weakly represented in titles and abstracts.

Taken together, the Results section should be read as a labeled-split evaluation plus submission diagnostics rather than a complete official TEST effectiveness report. Our strongest conclusions are those directly supported by per-query gold labels and formal inference (paired Wilcoxon tests with Holm correction) on the TRAIN and DEV splits; where post-submission TEST gold labels were later obtained, those labeled TEST analyses are also used to support claims in this report. Results based solely on pre-label submission tables should be interpreted as tentative diagnostic evidence rather than definitive effectiveness claims.

In Figure 4, a visual representation of the distribution is shown.

5.6.5. Error patterns and failure analysis

Manual inspection of missing and low-ranked cases suggests several recurring failure modes (non-exhaustive and overlapping):

1. **Claim-to-metadata specificity mismatch:** Claims that emphasize fine-grained experimental results, numeric outcomes, or methods not reflected in titles/abstracts are hard for the retriever to match.
2. **Translation artifacts in non-English queries:** Literal or imperfect translations introduce term shifts that reduce lexical overlap and semantic alignment with the English corpus.
3. **Implicit references with little lexical overlap:** Queries phrased as vague or conversational references (“this study”) lack the lexical anchors that support retrieval, requiring stronger semantic modeling or additional context.

5.6.6. Performance by query characteristics

Figure 5 reports a query-level characterization of the Nemotron Baseline run on the English TRAIN split. Contrary to our initial intuition, token length alone is not positively associated with higher MRR@5 in this run: short queries with at most 20 tokens reach 0.760 MRR@5, medium-length queries reach 0.713, and long queries with at least 35 tokens reach 0.672. This suggests that longer tweet-like claims often add conversational or contextual material that does not necessarily help identify the target paper.

The named-entity cue analysis is more consistent with the retrieval intuition. Using a lightweight heuristic based on author-like capitalization patterns, acronyms, and institution or venue keywords, entity-rich queries reach 0.733 MRR@5, while entity-sparse queries reach 0.664. These observations remain exploratory and should be interpreted as heuristic patterns rather than formal statistical findings, but they support the value of hybrid retrieval that combines lexical anchors with semantic matching.

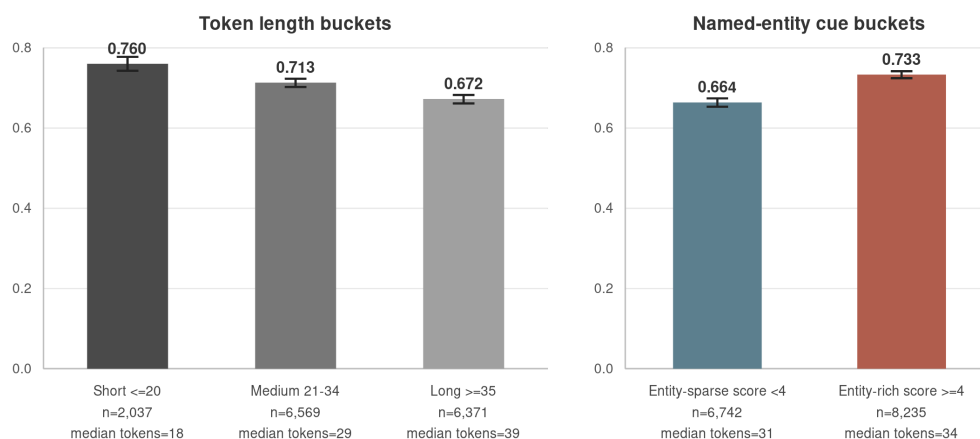


Figure 5: Mean per-query MRR@5 for the Nemotron Baseline English TRAIN run grouped by query length and by a lightweight named-entity cue score. Error bars show normal-approximation 95% confidence intervals over queries.

5.7. Reproducibility and summary

The analysis scripts used for descriptive summaries and inference are in `code/py/`, and the per-metric CSV artifacts are under `results/statistics/`.

In summary: combining descriptive summaries, stage-aligned formal inference, and rank-bucket diagnostics yields a coherent picture: H513 is the most defensible hybrid candidate generator on Recall@100, and Nemotron reranking (especially N1000) provides large, statistically supported gains in early precision (MRR@5). Remaining errors split between retrievable but low-ranked cases and true retrieval misses; addressing the latter requires improved candidate coverage or query reformulation.

6. Conclusions and Future Work

The results of the described system support three main conclusions.

First, analyzer and BM25 tuning provide a useful lexical baseline, especially thanks to its low computational footprint, with the best analyzer-focused run reaching 0.503 MRR@5 and 0.796 Recall@100 in the English run analysis.

Second, moving to BGE-M3 hybrid retrieval produces a stronger candidate generator: the retained configuration, `bge_m3_hybrid_513`, reaches 0.557 MRR@5 and 0.839 Recall@100.

Third, reranking the hybrid candidates with `nvidia/llama-nemotron-rerank-1b-v2` gives the largest observed improvement among the reported single-stage runs, reaching 0.702 MRR@5 and 0.911 Recall@100 when reranking the top 1000 candidates. Alternative rerankers such as `Qwen3-reranker-4B` and `Alibaba-NLP/gte-reranker-modernbert-base` also improve results, but remain below the Nemotron runs in the reported MRR@5 values.

The development results show that fine-tuning the Nemotron reranker is promising. Fine-tuning improves performance by a small but appreciable amount (an average gain of 0.03 MRR@5 points in our experiments).

Several directions emerge for future improvement. First, analyzing and spending more time for a better fine-tuning of the reranker and all other components of the system could bring better results. A lot of techniques like data augmentation can be experimented and there is room for improvement.

A second direction concerns the queries that remain entirely missing after reranking. These cases cannot be recovered by any reranker because the correct publication is absent from the candidate set. Manual inspection of a sample suggests three dominant failure patterns: claims referencing quantitative findings not mentioned in the abstract, implicit references that name no concrete entities and translation drift that changes the intended meaning of domain-specific terms. Addressing the first two patterns would probably require to manage content beyond title, abstract and authors but also other metadata. Addressing the translation failure pattern points toward a dedicated post-editing step or a domain-adapted translation model fine-tuned on biomedical text.

Finally, the German dataset achieves MRR@5 of 0.68 in the evaluation results: a wide gap relative to English and French. Controlled ablations that isolate translation errors from retrieval errors would clarify how much of this gap is attributable to the translation stage versus vocabulary mismatch in the retrieval stage itself.

7. Declaration on Generative AI

During the writing of this paper the authors used Grammarly and DeepL for grammar and spelling checks and Claude Sonnet 4.6 for writing-style improvement, citation management, and formatting assistance. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the publication content.

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